

Stepwise Ontogenetic Reversal and Structural Aging:

A Markov Framework for Local Cellular Plasticity

Abstract

Cellular plasticity in multicellular organisms is typically described through epigenetic landscapes, regulatory networks, or continuous developmental trajectories. However, these approaches rarely address whether reversibility itself is structurally constrained.

In this work, we propose a conceptual framework in which ontogenetic states form a locally connected Markov chain with predominantly adjacent transitions. Cellular reversal is interpreted not as unrestricted dedifferentiation, but as a stepwise and cascade-limited process constrained by structural organization, tissue compatibility, and temporal dynamics.

We introduce discrete ontogenetic states S_n , where S_0 denotes the terminal differentiated state and S_n progressively earlier developmental configurations. We further define: (i) a structural transition time τ_n ; (ii) a tissue compatibility space S^T ; and (iii) a plasticity parameter ϕ , representing the residual contribution of the immediately preceding developmental state.

The model provides a unified interpretation for fetal hemoglobin reactivation in maternal erythropoiesis, lineage-restricted regeneration, partial cellular plasticity, and the instability of deep reprogramming states. A novel prediction is advanced: elevated maternal HbF prior to or in early pregnancy reflects high ϕ in the maternal organism and constitutes a predisposition to impaired placentation and preeclampsia, independent of primary placental pathology.

1. Introduction

Differentiation in multicellular organisms is generally regarded as a stabilizing process in which cells progressively lose developmental plasticity. Nevertheless, multiple biological phenomena demonstrate that differentiated cells can partially regain characteristics associated with earlier developmental stages.

These include:

- reactivation of fetal gene programs,
- regeneration-associated dedifferentiation,
- metabolic reversion toward glycolysis,
- experimentally induced partial reprogramming [2].

Existing frameworks, most notably the epigenetic landscape model introduced by Waddington [1], describe these processes as movement across a continuous developmental surface. However, such models do not explicitly address whether ontogenetic reversibility is structurally constrained or whether intermediate states are obligatory.

In this article, we propose that ontogenetic reversal occurs predominantly through local transitions between adjacent developmental states, and that deeper reversals are progressively unstable due to incompatibility with the mature tissue environment.

2. Ontogenetic State Space

We define a discrete set of ontogenetic states:

$$S = \{ S_0, S_{-1}, S_{-2}, \dots, S_{-N} \}$$

where S_0 denotes the terminal differentiated state, S_{-1} the immediately preceding state, and $S_{-2} \dots S_{-N}$ progressively deeper developmental configurations. The ontogenetic axis is local rather than absolute: the same index k corresponds to lineage-specific positions rather than universal embryonic coordinates.

3. Local Transition Principle

Cellular transitions occur predominantly between adjacent ontogenetic states:

$$S_{-k} \leftrightarrow S_{-k-1}$$

while non-local transitions such as $S_0 \rightarrow S_{-2}$ are strongly suppressed. This defines ontogenesis as a locally connected state system rather than a freely traversable continuum.

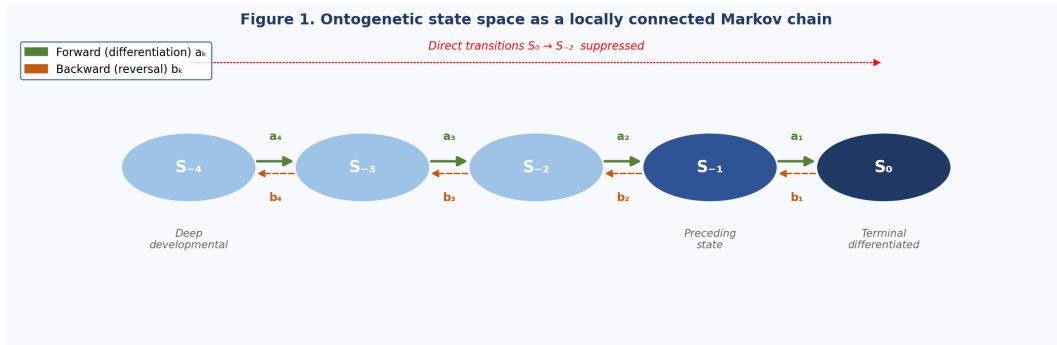


Figure 1. Ontogenetic state space as a locally connected Markov chain. Forward arrows (green) represent differentiation rates a_k ; backward arrows (orange, dashed) represent reversal rates b_k . Non-adjacent transitions (red dotted) are strongly suppressed.

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4. Markov Representation of Ontogenesis

We describe ontogenetic dynamics as a continuous-time Markov chain with forward transition rates (differentiation):

$$q_{-k, -k-1} = a_k$$

and backward transition rates (local reversal):

$$q_{-k}, q_{-k-1} = b_k$$

where a_k denotes differentiation rates and b_k denotes reversal rates. The local reversibility condition implies:

$$P(S_k \rightarrow S_{k-1}) \geq P(S_{k-1} \rightarrow S_k), \quad m > 1$$

5. Cascade-Limited Reversal

Deeper ontogenetic reversal can occur only through sequential intermediate transitions:

$$S_0 \rightarrow S_{-1} \rightarrow S_{-2} \rightarrow \dots$$

Each additional step is associated with decreasing transition probability, increasing structural instability, and declining population size. We propose exponential attenuation:

$$N(S_{-k}) \propto e^{(-\alpha k)}$$

where $N(S_{-k})$ is the population occupying state S_{-k} and α is a lineage-specific attenuation coefficient.

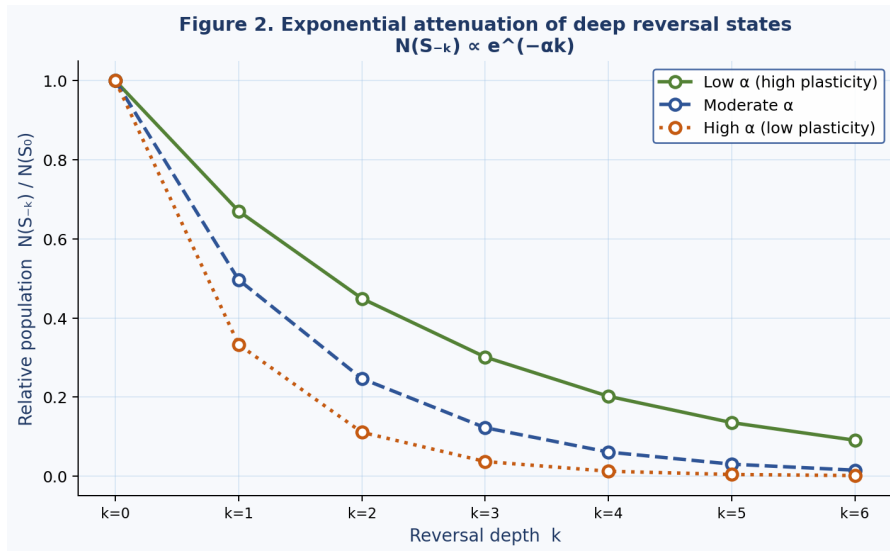


Figure 2. Exponential attenuation of deep reversal states $N(S_{-k}) \propto e^{-\alpha k}$ for three values of α . High α corresponds to aged or highly differentiated tissues; low α to embryonic or regenerative contexts.

Figure 2. Exponential attenuation of deep reversal states for three values of α . High α (low plasticity) corresponds to aged or highly differentiated tissues; low α (high plasticity) to embryonic or regenerative contexts.

6. Tissue-Compatible and Non-Compatible States

We distinguish tissue-integrated states $S_{-k}T$ and non-integrated states $S_{-k}U$. Transitions within the tissue-compatible subspace $S_{-k}T \rightarrow S_{-k-1}T$ remain structurally viable, whereas transitions into $S_{-k}U$ lead to loss of tissue integration and progressive instability. Viable ontogenetic reversal is

therefore constrained to the tissue-compatible subspace.

7. Structural Determination of Time

We introduce a temporal parameter $\tau_{\mathbf{S}}$, defined not as chronological age but as the characteristic structural transition time associated with state $\mathbf{S}$:

$$\tau_{\mathbf{S}} = f(\text{structure of } \mathbf{S})$$

where structure encompasses regulatory network organization, chromatin architecture, tissue integration, and intercellular coupling. Early developmental states have $\tau_{\mathbf{S}} \ll \tau_0$, while terminal differentiated states exhibit large τ_0 corresponding to structural inertia. Time emerges as a derived property of structural organization.

8. Plasticity Parameter ϕ

We define the residual contribution of the preceding ontogenetic state as:

$$\phi = S_{-1} / (S_0 + S_{-1})$$

High ϕ corresponds to elevated plasticity; low ϕ to terminal stabilization. We hypothesize that ϕ decreases monotonically with organismal aging, reflecting progressive structural consolidation.

9. Aging as Structural Drift

Aging is interpreted as a coordinated shift in three coupled quantities:

$$S_0 \uparrow \phi \downarrow \Delta E_{\text{window}} \downarrow$$

Dominance of S_0 increases progressively, the residual contribution of S_{-1} diminishes, and the energetic window narrows. The consequence is a shift from plasticity toward structural stability: regenerative capacity declines, structural transitions slow, and resistance to acute perturbations increases. This formulation is consistent with the epigenetic drift hypothesis of Yang et al. [3], while offering a complementary discrete-state interpretation.

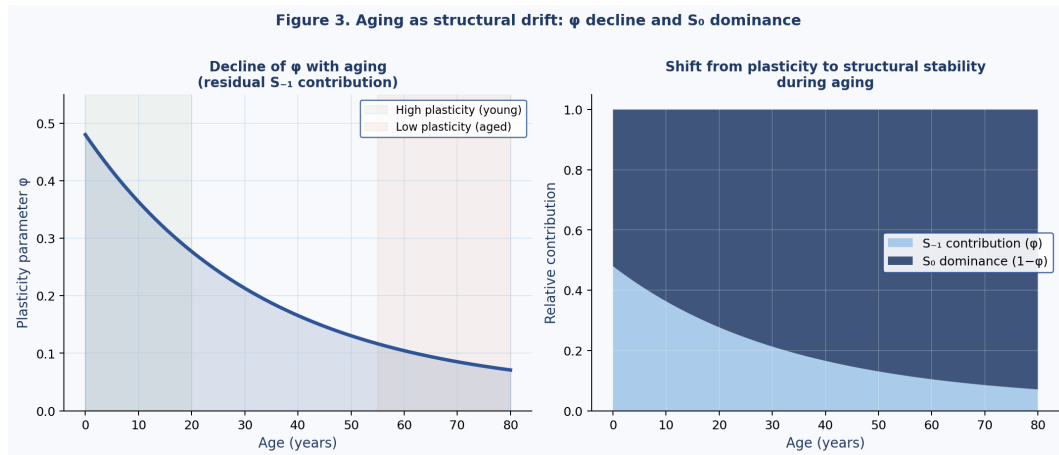


Figure 3. Aging as structural drift. Left: monotonic decline of ϕ with age. Right: progressive shift from S_{-1} contribution to S_0 dominance.

Figure 3. Aging as structural drift. Left: monotonic decline of ϕ with age, with high-plasticity (green) and low-plasticity (orange) zones indicated. Right: stacked area showing progressive shift from S_{-1} contribution to S_0 dominance.

10. Biological Examples

10.1 Fetal Hemoglobin in Maternal Erythropoiesis

HbF persists in small but measurable quantities in adult erythropoiesis, representing a local ontogenetic contribution of state S_{-1} to the terminally differentiated erythroid lineage S_0 :

$$S_0 \leftarrow S_{-1} (\phi > 0)$$

The level of HbF in adult erythrocytes is a direct quantitative indicator of ϕ in the erythropoietic lineage. High maternal HbF therefore reflects an organism-level state of incomplete ontogenetic stabilization — a property of the maternal organism prior to and independent of pregnancy. Elevated maternal HbF is not caused by pregnancy; it is a pre-existing characteristic of the erythropoietic state space of that individual.

10.2 Maternal HbF and Preeclampsia: A Novel Prediction

Multiple independent studies have reported elevated HbF in the serum of women who subsequently develop preeclampsia, detectable already in the first trimester of pregnancy [4–7]. The standard interpretation attributes this elevation to fetal hemoglobin of placental origin leaking through an oxidatively damaged blood–placenta barrier [4, 7].

We propose an alternative and mechanistically upstream interpretation: the HbF measured in early pregnancy may be substantially influenced by maternal erythropoietic HbF production rather than arising exclusively from fetoplacental leakage. This interpretation is supported by three converging observations:

- HbF differences between preeclamptic and control groups are detectable before overt clinical manifestations of placental pathology [5, 6].
- Elevated HbF has been identified in maternal erythrocytes independently of acute placental disruption [12].

- A quantitative mass-balance argument (see Section 10.2.1) suggests that fetoplacental leakage alone may be insufficient to account for observed first-trimester concentrations.

10.2.1 Quantitative Mass-Balance Consideration

At approximately 13 weeks of gestation, fetal mass is roughly 70–75 g, corresponding to an estimated fetoplacental blood volume of 9–12 mL and a total fetal hemoglobin pool of approximately 0.8–1.0 g Hb. Elevated cfHbF concentrations reported in preeclamptic women, when converted to total HbF mass across maternal plasma volume (~3 L), yield quantities approaching 10% of the estimated total fetal hemoglobin pool at this gestational age. While fetoplacental leakage cannot be excluded as a contributing source, this calculation suggests it may be insufficient as a sole explanation, supporting evaluation of maternal erythropoietic HbF production as an additional or primary source.

10.2.2 Mechanistic Prediction

High maternal HbF $\rightarrow$ high ϕ $\rightarrow$ incomplete ontogenetic stabilization of maternal tissues

We hypothesize that women with elevated maternal HbF prior to or during early pregnancy exhibit high ϕ , implying that the tissue environment required for normal trophoblast invasion and spiral artery remodeling is insufficiently mature structurally. This predisposition to impaired placentation arises independently of and prior to placental pathology.

This prediction is:

- Falsifiable: maternal HbF can be measured before conception or in the first trimester before clinical signs appear, and compared to known markers such as PAPP-A and PIGF.
- Mechanistically distinct: existing literature treats elevated cfHbF as a downstream marker of placental oxidative damage [4, 7, 11]; this framework predicts it as an upstream maternal predisposing state.
- Clinically actionable: pre-conceptional HbF measurement could identify women at elevated risk, enabling earlier preventive intervention.

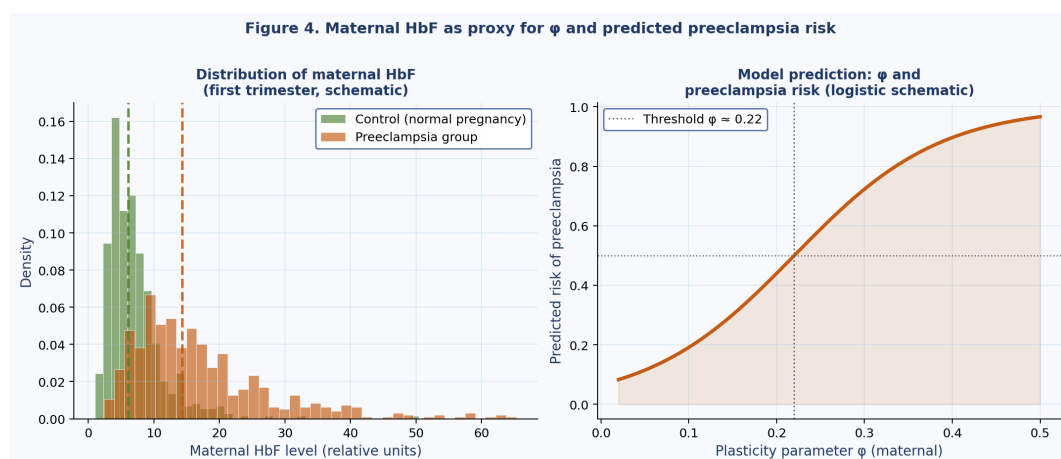


Figure 4. Maternal HbF as proxy for ϕ and predicted preeclampsia risk. Left: schematic HbF distributions (first trimester). Right: logistic model relating ϕ to preeclampsia risk.

Figure 4. Maternal HbF as proxy for plasticity parameter ϕ and predicted preeclampsia risk. Left: schematic HbF distributions in control and preeclamptic groups (first trimester). Right: logistic model relating ϕ to predicted preeclampsia risk, with threshold indicated.

10.3 Regeneration

In regenerative systems, particularly in amphibians, cells characteristically revert to lineage-restricted precursor states rather than pluripotency. This corresponds to local transitions $S_0T \rightarrow S_{-1}T$, consistent with the cascade-limited reversal principle, with deeper transitions remaining exponentially rare.

11. Discussion

The proposed framework differs from continuous epigenetic landscape models [1] by introducing four structural constraints: discrete local reversibility, cascade dependence, structural time, and tissue compatibility boundaries. Together these generate a model in which ontogenetic reversibility is not a freely available resource but a structurally limited capacity that diminishes with differentiation and age.

The model generates the following testable predictions:

- Deep reversal states should be exponentially rare, with frequency decreasing as $e^{(-\alpha k)}$.
- Regenerative potential should correlate positively with residual S_{-1} contribution (ϕ).
- Aging should correspond to increasing structural inertia (τ_0) and decreasing ϕ .
- Deep reprogramming without a permissive tissue environment should remain unstable.
- Elevated maternal HbF before or in early pregnancy should be associated with increased risk and severity of preeclampsia, independent of placental pathology markers.

The reinterpretation of first-trimester maternal HbF as a marker of the maternal ontogenetic state, rather than exclusively a product of placental damage, constitutes the principal novel and clinically actionable prediction. Existing first-trimester biobanks [5, 6, 9, 10] may contain the data necessary for a retrospective test of this hypothesis.

12. Conclusion

We propose that ontogenesis can be represented as a locally connected Markov process with structurally constrained reversibility. Within this framework:

- differentiation is locally reversible but cascade-limited in depth,
- aging reflects progressive dominance of terminal structural states and reduction of ϕ ,
- regenerative decline emerges from reduction of residual developmental plasticity,
- elevated maternal HbF constitutes a systemic marker of high ϕ and predicts structural predisposition to impaired placentation.

The model provides a conceptual bridge between development, regeneration, aging, and maternal predisposition to pregnancy complications. The specific prediction concerning maternal HbF and preeclampsia is falsifiable by prospective measurement of HbF prior to conception or in the early first trimester, and is mechanistically independent of the oxidative-damage hypothesis currently dominant in the literature.

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